AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

EFFLUENT TREATMENT PLANT (ETP)

MBBR TECHNOLOGY

CAPACITY RANGE : 10 KLD TO 500 KLD | OPERATION HOURS : 22 HRS

- Effective removal of BOD, COD, TSS, Oil & Grease, TDS (as required)
- Primary & Secondary Clarifier with MBBR
- Suitable for wide range of industrial effluents
- Customized design as per industry and discharge norms
- Compact, robust and reliable design
- Available in Civil / Package (MSEP) / Containerized options

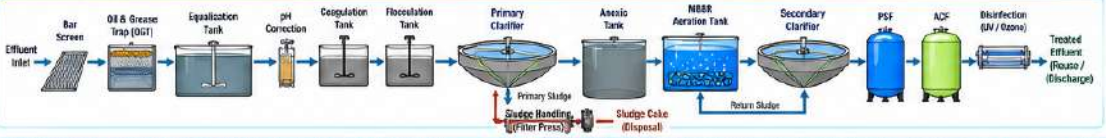
SUSTAINABLE TREATMENT FOR A CLEANER TOMORROW



INDUSTRIES & SECTORS WE SERVE



PROCESS FLOW DIAGRAM (ETP - MBBR TECHNOLOGY)



TYPICAL INLET & OUTLET PARAMETERS (INDUSTRIAL EFFLUENT)

Parameter	Inlet (Typical Range)	Outlet (Typical)
pH	5.5 – 9.0	6.5 – 8.5
BOD (mg/L)	200 – 1500	< 30
COD (mg/L)	500 – 3000	< 100 (or as per requirement)
TSS (mg/L)	100 – 1000	< 50
Oil & Grease (mg/L)	10 – 100	< 10
TDS (mg/L)	500 – 5000 (depending on industry)	As per requirement
Ammonia – N (mg/L)	10 – 100	< 10
Color (Pt-Co)	50 – 500	< 20
Heavy Metals (mg/L)	As per industry	As per discharge norms

Note: Values are indicative and may vary based on effluent characteristics, applicable discharge norms and client requirements.

TECHNICAL SPECIFICATIONS – 10 KLD TO 500 KLD (OPERATION HOURS : 22 HRS)

Parameter	10 KLD	25 KLD	50 KLD	100 KLD	200 KLD	300 KLD	500 KLD
Design Flow (KLD)	10	25	50	100	200	300	500
Average Flow (m³/hr)	0.45	1.14	2.27	4.55	9.09	13.64	22.73
Hydraulic Retention Time (hrs)	8 – 12	8 – 12	8 – 12	8 – 12	8 – 12	8 – 12	8 – 12
MBBR Media Fill (%)	40 – 50	40 – 50	40 – 50	40 – 50	40 – 50	40 – 50	40 – 50
MBBR Media Quantity (m³)	1.5 – 2	3 – 4	6 – 8	12 – 16	24 – 32	36 – 44	60 – 80
Air Requirement (Nm³/hr)	20 – 30	40 – 60	80 – 120	150 – 200	300 – 400	500 – 600	800 – 1000
Blower Capacity (1 + 1)	2.5 HP	5 HP	10 HP	15 HP	25 HP	40 HP	60 HP
Connected Load (kW) (Approx.)	8 – 12	15 – 20	25 – 35	45 – 60	80 – 120	120 – 160	200 – 250
Total Area Required (m²) (Approx.)	35 – 50	60 – 80	90 – 120	150 – 200	250 – 350	350 – 450	500 – 700

MOC (MATERIAL OF CONSTRUCTION)

Item	Material
Civil Tanks (Optional)	RCC / MSEP (MS with FRP / Epoxy)
Package Plant (MSEP)	MS with FRP / Epoxy Lining
Piping (Process)	UPVC / CPVC / GI / SS-304 (as per service)
Pumps	CI / SS-304
Blowers	Root Blower / Turbo Blower
Diffusers	EPDM Fine Bubble Diffuser
MBBR Media	HDPE
Screens	SS-304
Clarifier (Primary & Secondary)	RCC / MSEP with FRP Lining
PSF & ACF Vessels (Optional)	FRP / MSEP / MSRL
Instrumentation	pH Meter, DO Meter, Flow Meter, Level Switch, CIP, etc.
Control Panel	MCC with PLC (Optional)

KEY EQUIPMENT (TYPICAL BOM)



OUR ETP EXECUTION OPTIONS



WHY AQUEN ?

Customized Design as per Industry | High Treatment Efficiency | Quality Equipment & Components | On-time Execution | Competitive Pricing | After Sales Support & AMC

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

Office & Factory:
Sr. No. 165/2/8, Alankaouram Road,
Opposite Yashobhumi Society, Adarsha Nagar,
Charholi Budruk, Pimpri-Chinchwad,
Maharashtra – 412105, India.

+91 9096645119 / +91 9622554445

ceo@aquenfluid.com

www.aquenfluid.com

1. ETP MBBR – 100 KLD (Civil Work in Progress)



2. ETP MBBR – 100 KLD (Electro – Mechanical Installation)



3. ETP MBBR – 250 KLD (Civil Work in Progress)



4. ETP MBBR – 250 KLD (Electro – Mechanical Installation)



5. Package ETP MBBR – 50 KLD (MSRL/MSEP Installation)



6. ETP MBBR – 300 KLD (MSEP Tanks with Filtration System)

